MIL-T-46072A(MR)
10 September 1971
SUPERSEDING
MIL-T-46072(MR)
10 December 1963

MILITARY SPECIFICATION

TUBE, ROUND, COPPER ALLOY NUMBERS 330, 331, 332, AND 370 (LEADED BRASS), SEAMLESS

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers 4 alloys of leaded brass tube (see 1.2.1).

1.2 Classification.

1.2.1 Alloys. Tube furnished under this specification shall be of one of the following copper alloy numbers: 330, 331, 332, or 370.

1.2.2 Tempers. Leaded brass tube shall be furnished in one of the following tempers:

Light anneal
Soft anneal
Drawn (general purpose)
Hard drawn
Drawn and stress relieved (for primer bodies)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Std. No. 146 - Tolerances for Copper and Copper Base Alloy Mill Products
Fed. Test Method Std. No. 151 - Metals; Test Methods
Fed. Std. No. 185 - Identification Marking of Copper and Copper Base Alloy Mill Products

FSC 4710

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

B 154 - Mercurous Nitrate Test for Copper and Copper Alloys
E 8 - Tension Testing of Metallic Materials
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 112 - Estimating Average Grain Size of Metals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Chemical composition. The material shall conform to the chemical requirements specified in table I.

TABLE I. *Chemical composition*

Copper alloy number	Copper, percent	Lead, percent	Iron, max., percent	Zinc, percent	Total other elements, max., percent
330	65.0–68.0	0.2–0.8	0.07	Remainder	0.50
331	65.0–68.0	.7–1.2	.06	Remainder	.50
332	65.0–68.0	1.8–2.0	.07	Remainder	.50
370	59.0–62.0	0.9–1.4	.15	Remainder	.50

3.1.1 Analysis shall be made regularly only for the elements specifically mentioned in table I except zinc. If, however, the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

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3.2 Mechanical properties and grain size.

3.2.1 Tensile and hardness requirements, drawn tube (except tube for primer bodies). Drawn tube (see 1.2.2), shall conform to the tensile and hardness requirements specified in table II.

TABLE II. *Tensile and hardness requirements, drawn tube*

Copper alloy number	Temper	Outside diameter, inch	Wall thickness, inch	Tensile strength, minimum, psi	Rockwell hardness [A] 30T, minimum
330, 331, 332	Drawn [B]	All	All	54,000	53
	Hard drawn [C]	Up to 1, incl.	0.020 to 0.120, incl.	66,000	70
		Over 1 to 2, incl.	.035 to .180, incl.		
		Over 2 to 4, incl.	.060 to .250, incl.		
370	Drawn [B]	All	All	54,000	55

[A] Rockwell hardness values apply only to tube having a wall thickness of 0.020 inch or over.

[B] Drawn temper is for general purposes and is most commonly used where there is no specific requirement for high strength on one hand and bending qualities on the other.

[C] Hard-drawn temper is used only where there is a need for tubing as strong as is commercially feasible for size indicated. For any combination of diameter and wall thickness not covered under hard-drawn temper, use values given for drawn temper.

3.2.2 Hardness requirements, drawn and stress relieved tube for primer bodies (copper alloy numbers, 330, 331, and 332). Tube shall have a minimum Rockwell hardness of 50 measured on the 30T scale.

3.2.3 Grain size and hardness requirements, annealed tube. Annealed tube (see 1.2.2), shall conform to the grain size and hardness requirements specified in table III.

TABLE III. *Grain size and hardness requirements, annealed tube*

Copper alloy number	Temper	Wall thickness, inch	Average grain size, mm		Rockwell hardness [A], maximum	
			Minimum	Maximum	30T	F
330, 331, 332	Soft anneal	Up to 0.030, incl.	0.025	0.060	40	....
		Over 0.030	.025	.060	....	80
330, 331, 332, 370	Light anneal	Up to 0.030, incl.	([B])	.035	60	....
		Over 0.030	([B])	.035	....	90

[A] The F Scale hardness values apply only to tube 0.030 inch and over in wall thickness; 30-T Scale hardness values apply to tube 0.024 inch and over.

[B] Although no minimum grain size is specified, the tube must be fully recrystallized.

3.2.4 Expansion test. Drawn and stress relieved tube for primer bodies and annealed tube shall be subjected to the expansion test specified in 4.5.4. When expanded to the requirements shown in table IV, the tube shall show no cracks or ruptures when examined with the unaided eye, including the use of corrective spectacles necessary to obtain normal vision.

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TABLE IV. *Expansion test requirements*

Temper	Outside diameter, inch	Expansion of outside diameter, percent
Annealed	¾ and under	20
Annealed	Over ¾	15
Drawn and stress relieved	All	10

3.3 Mercurous nitrate test.

3.3.1 For primer bodies only. The drawn and stress relieved tube shall withstand the mercurous nitrate test specified in 4.5.6 without cracking.

3.3.2 Tube other than for primer bodies. When specified (see 6.2), both the light annealed and the drawn tempers shall withstand the mercurous nitrate test when tested in accordance with 4.5.6.

3.4 Dimensional tolerances. Unless otherwise specified in the drawing or in the contract or order, all tube shall meet the requirements of the following references of Fed. Std. No. 146.

Dimension	Reference
Diameter	20a(1)
Wall thickness	20a(2)
Length	20a(3), 20a(4)
Straightness	20a(5)
Squareness of cut	20a(6)
Roundness	20a(7)

3.5 Identification marking. When specified in the contract or order, product identification marking shall be in accordance with Fed. Std. No. 185. (see 6.2 and 6.3).

3.6 Workmanship. The tube shall be uniform in quality and temper, and shall be sound, clean, round, smooth, and with ends cut square. The tube shall be free from laminations, cracks, seams, scale, laps, pits, indentation, grooves and injurious defects. The annealed tube shall be either clean annealed or acid cleaned after final annealing operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or

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any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order, a lot of tube shall consist of 5,000 pounds or fraction thereof, of material of the same size, alloy, and temper.

4.3 Sampling.

4.3.1 Sampling for chemical analysis. Two lengths of tube shall be selected at random from each lot. From each length, not less than two ounces of clean metal, millings, clippings, or drillings shall be obtained.

4.3.2 Sampling for mechanical properties.

a. Tension test. Two specimens shall be selected from each lot of drawn tube, one each from different tubes, for determination of tensile strength. If the lot consists of only one tube, both specimens shall be taken from that tube.

b. Hardness test. Hardness determinations shall be made on five samples from each lot of tube.

c. Expansion test. At least two tubes shall be selected from each lot of drawn and stress relieved tube (for primer bodies) and of annealed tube for the expansion test.

4.3.3 Sampling for grain size determination. Three samples shall be selected for grain size determination from each lot of annealed tube.

4.3.4 Sampling for mercurous nitrate test. When required, one specimen at least 6 inches in length shall be selected from each lot of tube for the mercurous nitrate test (see 3.3).

4.3.5 Sampling for visual and dimensional examination.

4.3.5.1 Visual examination. From each lot of material, a representative sample of material shall be selected in accordance with inspection level III, MIL-STD-105, acceptable quality level (AQL) of 1.5 percent, and shall be visually examined as specified in 4.4.1.

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4.3.5.2 Dimensional examination. From each lot of material, a representative sample shall be selected in accordance with inspection level III, MIL-STD-105, acceptable quality level (AQL) of 1.5 percent, and shall be measured as specified in 4.4.2. The samples selected for dimensional examination may be the same as those selected for visual examination.

4.4 Examination.

4.4.1 Visual. Each sample unit selected in accordance with 4.3.5.1 shall be visually examined to determine compliance with the requirements for identification marking (see 3.5) and workmanship (see 3.6).

4.4.2 Dimensional. Each sample unit selected in accordance with 4.3.5.2 shall be measured to determine compliance with the dimensional requirements (see 3.4) of this specification. Straightness of each sample unit shall be determined as specified in 4.4.3.

4.4.3 Straightness. Straightness shall be determined by placing the piece on a level surface so that the arc or departure from straightness is horizontal. The maximum depth of arc shall be measured to the nearest 1/32 inch by means of a straightedge and a steel scale.

4.4.4 Preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Tests.

4.5.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151. In case of dispute, analysis by method 111 shall be the basis for acceptance or rejection.

4.5.2 Tension tests. Specimens for tensile testing shall be prepared and tested in accordance with ASTM E 8.

4.5.3 Hardness tests. Specimens for hardness testing shall be prepared and tested in accordance with ASTM E 18.

4.5.4 Expansion test. The test specimen shall be of suitable length, at least twice the outside diameter of the tube. Both ends of the specimen shall be faced square on a lathe, or suitably prepared so as to have a smooth surface free from scratches or burrs, with both edges slightly chamfered. The inside surface of the specimen shall be well lubricated at the test end with No. 1 lard oil or any extreme-pressure lubricating oil. The test specimen shall be expanded over a hardened and ground tapered steel pin, having an included angle of 60°, until the expansion specified in table IV is obtained. The pin, which also shall be lubricated, must have a diameter suitable for the size of the tube and the amount of expansion required. During the expansion, the testing machine shall be operated at a rate not to exceed a crosshead speed of 2 inches per minute. The longitudinal axes of the test specimen and the pin shall coincide.

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4.5.5 Grain size determination. Specimens for grain size testing shall be prepared and tested in accordance with ASTM E 112.

4.5.6 Mercurous nitrate test. Mercurous nitrate tests shall be conducted in accordance with ASTM B 154.

4.6 Rejection.

4.6.1 Examination defects. Any sample tube having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.5.1 and 4.3.5.2 for that sample size, the entire lot shall be rejected subject to the section on "Acceptance and Rejection" of MIL-STD-105.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5, subject to the rejection and retest provisions of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing, level C. Material covered by this specification shall be packaged and packed for shipment in compliance with common carrier regulations applicable to that mode of transportation to ensure safe delivery at destination at lowest transportation cost without assessment of penalty charges for improper packing (see 6.2 and 6.3).

5.2 Marking. In addition to any special marking required in the contract or order (see 6.2 and 6.3), marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as follows:

a. Copper alloy numbers 330, 332, and 370 - Fire control equipment and similar applications requiring a free-machining material.

b. Copper alloy numbers 330, 331, and 332 - Primer bodies for artillery ammunition.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Alloy, temper, and size required (see 1.2).

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c. When tube is required for primer bodies (see 3.2.2 and 3.3.1). In such cases, alloy numbers 330, 331, and 332 may be ordered as a group.

d. When mercurous nitrate test for tube other than for primer bodies is required (see 3.3.2).

e. When product identification marking in accordance with Fed. Std. No. 185 is required (see 3.5).

f. When a lot size larger than 5,000 pounds is required (see 4.2).

g. Maximum gross weight of container (see 5.1).

h. Special marking, if required (see 5.2).

6.3 The requirements for product identification marking (see 3.5) and for packing (see 5.1) and marking (see 5.2) for shipment specified herein apply to direct shipment for Government activities and apply also, when specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 The chemical composition and mechanical properties of the material covered by this specification are similar to the chemical composition and mechanical properties of the following ASTM alloys:

MIL-T-46072A(MR) Alloy	ASTM B 135-71 Alloy
330	3
331	-
332	4
370	6

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - MU
DSA - CS

Project No. 4710-A0207

User activities:
Army - MI

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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